DSA0503 - Query Processing for Data Science Activity

Name : Anto Chriswin G

Register Number : 192324093

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1) Employee and Department Database Design (10 Marks)

A company wants to store employee information and department details in an SQLite database. Design two related tables, Department and Employee, by identifying appropriate primary and foreign keys. Write a Python program to connect to the database, create both tables, insert sample data into each table, and display the employee name along with the corresponding department name using an SQL JOIN query.

```python
import sqlite3

# Connect to the SQLite database
conn = sqlite3.connect("company.db")
cursor = conn.cursor()

# Create Department table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Department(
    Department_ID INTEGER PRIMARY KEY,
    Department_Name TEXT NOT NULL
)
""")

# Create Employee table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Employee(
    Employee_ID INTEGER PRIMARY KEY,
    Employee_Name TEXT NOT NULL,
    Salary REAL,
    Department_ID INTEGER,
    FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)
)
""")

# Insert sample data into Department table
departments = [
    (1, "HR"),
    (2, "Finance"),
    (3, "IT")
]
```

```python
cursor.executemany(
    "INSERT INTO Department (Department_ID, Department_Name) VALUES (?, ?)",
    departments
)

# Insert sample data into Employee table
employees = [
    (101, "John", 50000, 1),
    (102, "Alice", 60000, 2),
    (103, "David", 70000, 3),
    (104, "Emma", 65000, 3)
]

cursor.executemany(
    "INSERT INTO Employee (Employee_ID, Employee_Name, Salary, Department_ID) VALUES (?, ?, ?, ?)",
    employees
)

# Execute INNER JOIN query
cursor.execute("""
SELECT Employee.Employee_Name,
       Department.Department_Name
FROM Employee
INNER JOIN Department
ON Employee.Department_ID = Department.Department_ID
""")

# Display the results
print("Employee Name\tDepartment")
print("-" * 30)

for row in cursor.fetchall():
    print(f"{row[0]}\t\t{row[1]}")

# Commit and close the connection
conn.commit()
conn.close()
```

Output :

```
PS F:\COURSES\QP-DS\Assessment Tool\co2_at1> python at1.py
Employee Name   Department
------------------------------
John            HR
Alice           Finance
David           IT
Emma            IT
```

2) Banking Customer Database (10 Marks)

A bank wants to maintain customer account details in MySQL. Design an Account table with appropriate constraints. Write a Python program to connect to the database, create the table, insert customer account records, update the account balance after a transaction, and retrieve the account information.

```python
import mysql.connector

# Connect to MySQL database
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="jeschris",
    database="antodb1"
)

cursor = conn.cursor()

# Create Account table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Account(
    Account_No INT PRIMARY KEY,
    Customer_Name VARCHAR(50) NOT NULL,
    Account_Type VARCHAR(20) NOT NULL,
    Balance DECIMAL(10,2) NOT NULL CHECK (Balance >= 0)
)
""")

# Insert customer account records
accounts = [
    (1001, "John", "Savings", 25000.00),
    (1002, "Alice", "Current", 18000.00),
    (1003, "David", "Savings", 32000.00)
]

cursor.executemany("""
INSERT INTO Account(Account_No, Customer_Name, Account_Type, Balance)
VALUES (%s, %s, %s, %s)
""", accounts)

# Update account balance after a transaction
cursor.execute("""
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 1001
""")

# Retrieve account information
cursor.execute("SELECT * FROM Account")

print("Account Details")
print("-" * 60)

for row in cursor.fetchall():
    print(row)

# Commit changes
conn.commit()

# Close connection
cursor.close()
conn.close()
```

Output :

```
(.venv) PS F:\COURSES\QP-DS\Assessment Tool\co2_at1> python at2.py
Account Details
------------------------------------------------------------
(1001, 'John', 'Savings', Decimal('30000.00'))
(1002, 'Alice', 'Current', Decimal('18000.00'))
(1003, 'David', 'Savings', Decimal('32000.00'))
```